

Project Proposal

What are you trying to do?



How is it done today, and what are the limits of current practice?

Related Work	What It Shows	Limitation
AI MockPrep	AI interview preparation systems can simulate interviews and provide feedback	Limited domain adaptability; advanced technical answers are harder to evaluate
Stamper et al.	LLMs can generate feedback through prompts	Feedback lacks strong design foundation and empirical evaluation
Shankar et al.	LLMs can assist evaluation	LLM evaluators still need human validation; criteria drift

Gap: Feedback quality is still hard to control and evaluate.

What is new in your approach and why do you think it will be successful?

- * Problem Domain: Existing (Interview Simulation)**
- * Approach: New Technique (RLAIF + N-wise Ranking)**
- * Key Differences: Upgrading from "Correctness" to "Expert Coaching Alignment"**
- * Why It Succeeds: Trains the model on how to be a helpful mentor, not just a grader**

Who cares? If you are successful, what difference will it make?

- * **Practical Motivation: Providing elite CS mentorship to every student**
- * **Global Change: Expert help for everyone, 24/7**
- * **New Capabilities: Role-Specific Alignment, Multidimensional Critique**
- * **System Benefits: Faster, Accurate, Interactive**

What are the risks and how long will it take?

Risks:

1. Feedback Quality

The model may give vague, incorrect, or overly general feedback.

2. Evaluation

It may be difficult to prove that the feedback is truly better than a baseline model.

3. Data:

Creating high-quality ranked feedback examples may take more time than expected.

What are the risks and how long will it take?

Timeline (about 5 weeks):

1. Dataset Building & Preparation (2 weeks)

- Collect questions, answers, and feedback
- Construct ranking / preference data

2. Model Training (2 weeks)

- Supervised fine-tuning
- RLAIIF / alignment training

3. Evaluation (1 week)

- Midterm + final metrics (LLM judge + human evaluation)

What are the midterm and final “exams” to check for success?

- **The midterm(after fine tuning)**

- 1. Technical Accuracy Scoring:**

- 2. Feedback Coverage Rate:**

- 3. Win Rate Calculation(LLM-as-a-Judge)**

What are the midterm and final “exams” to check for success?

- **The final exam(after RLHF)**

- 1.Human-Model Agreement Rate**

- 2. Usefulness Score (Likert Scale)**

- 3. Win Rate Calculation(human evaluation)**

Reference

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